

03. Hormonal Contraception in Medical Comorbidity

Companion reading handout for simultaneous listening.

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The most dangerous way to learn contraceptive eligibility is to memorize isolated contraindications.

It produces two different kinds of harm.

One resident gives estrogen to a patient with a true category-four risk, because she seems young and healthy.

Another resident denies effective contraception to a patient because a label sounds frightening, even though the label is not actually a contraindication.

Both errors come from the same weak mental model.

They treat eligibility as a list of scary words.

The better model is this:

First name the method.

Then name the risk mechanism.

Then name the category.

Then offer the safer effective alternative.

That sequence matters.

If you skip the method, you may forget that combined pills, patch, and ring behave differently from progestin-only pills, implant, DMPA, copper IUD, and levonorgestrel IUD.

If you skip the mechanism, you will confuse MTHFR with antiphospholipid antibodies, migraine without aura with migraine with aura, or stable renal transplant with graft failure.

If you skip the category, you will treat category three and category four as if they are the same.

And if you skip the alternative, counseling becomes a prohibition instead of care.

The organizing tool is the US MEC: the US Medical Eligibility Criteria for Contraceptive Use.

ACOG's hormonal contraception guidance teaches the key intellectual move. The US MEC does not compare a method with zero risk. It compares method risk with the risk

of unintended pregnancy in the context of that patient's condition.

That is why contraception in comorbidity matters so much.

Hypertension, diabetes, valvular disease, ischemic heart disease, stroke, teratogenic medications, transplant status, epilepsy, and cancer history do not only change contraceptive risk. They also change the risk of pregnancy.

So the question is not, is there any risk?

There is always some risk.

The question is: which risk is acceptable, and which method gives reliable pregnancy prevention without adding avoidable danger?

The categories give language to that balance.

Category one means no restriction.

Category two means advantages generally outweigh theoretical or proven risks.

Category three means risks usually outweigh advantages. The method is usually not recommended, but it may be used if better options are not acceptable or available, with counseling and follow-up.

Category four means unacceptable health risk.

Do not flatten those categories.

Category three is not "absolutely forbidden."

Category four is not "use if convenient."

Place estrogen at the center of the map

Combined hormonal contraceptives include pills, patch, and vaginal ring. The estrogen component increases hepatic production of coagulation-related proteins, including factor seven, factor ten, and fibrinogen. That is why combined hormonal contraception increases venous thromboembolism risk.

But this must be taught precisely.

ACOG notes that the VTE risk with combined hormonal contraception remains lower than the VTE risk observed in pregnancy. That does not mean estrogen is safe for everyone. It means the resident must compare three things: the risk of the method, the risk of unintended pregnancy, and the availability of safer effective alternatives.

So the first estrogen question is not, does estrogen have risk?

It does.

The first estrogen question is, does this patient have a condition that makes estrogen's clotting, vascular, or stroke risk unacceptable?

VTE

A patient with prior VTE should generally be counseled toward nonhormonal or progestin-only contraception. The clean alternatives are often copper IUD, levonorgestrel IUD, implant, or progestin-only pills, depending on the rest of the clinical picture.

But do not oversimplify prior VTE.

The recurrence risk depends on whether the original clot was connected to a permanent risk factor, such as factor five Leiden, or to a reversible nonrecurring trigger, such as surgery or temporary immobilization years ago.

ACOG gives an important nuance: a combined hormonal method may be considered as category three in a patient with a single remote VTE associated with a nonrecurring risk factor, if no other method is acceptable or available.

That is not first-line. It is not casual. It is category three.

The resident reflex is: prior VTE usually points away from estrogen, but category three is a risk-negotiation category, not a magic eraser.

Compare that with known thrombophilia

Known familial thrombophilias make combined hormonal contraception category four. This includes conditions such as factor five Leiden, prothrombin mutation, protein C deficiency, protein S deficiency, antithrombin deficiency, and antiphospholipid syndrome.

Here the mechanism is not vague. The patient already has a thrombogenic biology. Estrogen adds hepatic coagulation effects on top of it.

Progestin-only methods and levonorgestrel IUD are acceptable alternatives in known thrombogenic mutations, generally category two.

But now comes a classic trap.

MTHFR is not the same thing.

ACOG states that MTHFR mutations alone do not appear to convey increased VTE risk, and assessment of MTHFR polymorphisms or fasting homocysteine is not recommended in this setting.

That means the exam and bedside contrast is not "thrombophilia sounding word equals no estrogen."

The contrast is real thrombogenic mutation versus MTHFR alone.

Antiphospholipid antibodies are real thrombotic risk.

MTHFR alone is not treated as the same disease.

The resident reflex is: a contraindication is not a vibe. It is a mechanism plus a category.

Add SLE

Systemic lupus erythematosus increases arterial and venous thrombosis risk. Antiphospholipid antibodies increase it further. ACOG recommends testing women with SLE for antiphospholipid antibodies before initiating hormonal contraception.

If antiphospholipid antibodies are positive or unknown, combined hormonal contraception is category four.

If SLE is present without antiphospholipid antibodies and without other cardiovascular risk factors, combined hormonal contraception is category two and does not appear to worsen inactive or stable active disease.

That distinction is clinically powerful.

The disease name alone is not enough.

The antibody status changes the answer.

The resident reflex is: in lupus, ask about antiphospholipid antibodies before you think about estrogen.

Smoking

Smoking is not one category.

Age changes the answer.

For combined hormonal contraception, smoking under age thirty five is not the same as smoking at thirty five or older. At age thirty five or older, fewer than fifteen cigarettes per day is category three. Fifteen or more cigarettes per day is category four.

So a twenty five year old smoker needs counseling about smoking, cardiovascular health, and contraception, but she is not the same eligibility problem as a thirty eight year old heavy smoker.

This is the larger lesson:

Do not prescribe estrogen because the patient is young.

But also do not deny estrogen because one risk word appears before you have applied the category.

Postpartum contraception

Postpartum estrogen risk is dynamic.

Regardless of breastfeeding status, combined hormonal contraception is category four in the first twenty one days after delivery. The reason is clotting physiology. The early postpartum period is hypercoagulable, and estrogen adds avoidable thrombotic risk.

From day twenty one to day forty two, the risk decreases but still depends on breastfeeding and VTE risk factors such as age, prior VTE, thrombophilia, immobility, transfusion, peripartum cardiomyopathy, body mass index of thirty or greater, postpartum hemorrhage, cesarean delivery, preeclampsia, or smoking.

After forty two days, nonbreastfeeding patients have no restriction for combined hormonal contraception.

Breastfeeding patients move a little differently. Progestin-only methods may be used at any time postpartum. The theoretical concern is that progesterone withdrawal after placental delivery helps trigger lactogenesis, so immediate progestin exposure raised concern about breastfeeding. But observational data do not show clear harm to breastfeeding initiation, continuation, infant growth, or infant development.

Combined hormonal contraception in breastfeeding is more cautious early, partly because of VTE and partly because lactation data are limited and inconsistent.

The memory line is this:

Early postpartum estrogen is a clotting problem first.

Breastfeeding is a separate counseling layer.

Migraine

Migraine without aura and migraine with aura are different contraceptive problems.

Progestin-only methods have no restriction in migraine, with or without aura.

Combined hormonal contraception can be used in migraine without aura if no other stroke risk factors are present, category two.

Migraine with aura makes estrogen-containing contraception category four.

The mechanism is arterial stroke risk. Aura is not just a bad headache. Aura is a focal reversible neurologic symptom complex, often visual, usually lasting about five to sixty minutes.

The resident must avoid two errors.

Do not call every headache aura.

Do not miss true aura.

If true aura is present, the estrogen answer changes.

Hypertension

Hypertension is where category thresholds teach risk physiology.

Blood pressure below one forty over ninety permits any hormonal method.

Systolic one forty to one fifty nine, or diastolic ninety to ninety nine, makes combined hormonal contraception category three.

Systolic one sixty or higher, diastolic one hundred or higher, or vascular disease makes combined hormonal contraception category four.

Controlled hypertension on medication is also category three, because treated blood pressure does not erase the underlying vascular-risk context. ACOG allows that a carefully selected healthy nonsmoking woman age thirty five or younger, with controlled hypertension and no end-organ vascular disease, may be considered for a trial with careful monitoring.

This is not a permission slip to ignore blood pressure.

It is a reminder that category three requires judgment, documentation, counseling, and alternatives.

Progestin-only pills do not appear to affect blood pressure significantly. DMPA is more cautious in severe hypertension because of theoretical lipid and cardiovascular concerns.

Diabetes

Uncomplicated diabetes does not contraindicate hormonal contraception.

But diabetes for more than twenty years, or diabetes with microvascular disease such as retinopathy, nephropathy, or neuropathy, changes the vascular-risk profile.

Combined hormonal contraception becomes category three or four depending on severity. DMPA also becomes more concerning in complicated diabetes because of lipid concerns.

Again, the disease label is not enough.

Diabetes without vascular disease is not the same as long-standing diabetes with microvascular disease.

Now solid-organ transplant, including renal transplant.

This is a common exam trap because many residents hear "renal transplant" and reflexively say no estrogen.

The CDC US MEC twenty twenty four is more precise.

Solid-organ transplantation is a condition in which pregnancy itself may carry increased risk. But contraceptive eligibility depends on graft status.

With no graft failure, combined hormonal contraception is category two.

With graft failure, combined hormonal contraception is category four.

For copper and levonorgestrel IUD methods, the CDC table separates initiation and continuation. With no graft failure, initiation and continuation are category one. With graft failure, initiation is category two and continuation is category one.

Implant and progestin-only pills are category two. DMPA is category two or three, depending on fracture risk and long-term immunosuppression context.

The mechanism is that transplant is not a single physiologic state. A stable graft and graft failure have different renal, vascular, hypertensive, metabolic, infectious, and medication contexts.

The resident reflex is: do not stop at "renal transplant." Ask whether there is graft failure.

Obesity and bariatric surgery

Obesity alone does not contraindicate hormonal contraception. ACOG states that women with obesity can be offered all hormonal method options, with reassurance that efficacy is not significantly affected by weight in available evidence.

But obesity and combined hormonal contraception both increase VTE risk, so the resident should still consider the whole patient, especially if other risk factors are present.

Bariatric surgery is different because it can change absorption.

After malabsorptive procedures such as Roux-en-Y gastric bypass or biliopancreatic diversion, oral contraception should not be used because efficacy may be impaired; this is category three. Nonoral methods are category one. Restrictive procedures such as sleeve gastrectomy do not create the same oral absorption concern.

The mechanism is not "bariatric surgery equals no hormones."

It is malabsorption equals unreliable oral exposure.

Depression

Women with depressive disorders can use all hormonal contraception methods, category one. Depressive symptoms do not appear to worsen with hormonal methods,

including DMPA. SSRI and SNRI medications do not appear to alter hormonal contraceptive metabolism.

But St. John's wort is different. It is a hepatic enzyme inducer and can increase steroid metabolism, breakthrough bleeding, and ovulation likelihood.

Again, the pattern repeats:

The label is not enough. The mechanism matters.

Older reproductive age

Healthy nonsmoking women without specific cardiovascular risk factors may continue combined hormonal contraception until age fifty to fifty five, category two. Age alone is not the contraindication. Age increases the chance that other contraindications appear.

FSH testing is not a reliable way to decide that a hormonal contraceptive user is menopausal and no longer needs contraception.

Combined hormonal contraception may help perimenopausal patients with abnormal bleeding, vasomotor symptoms, bone mineral density, and reduced ovarian and endometrial cancer risk.

But be precise. The source does not teach combined pills as reducing breast cancer risk.

Cancer history

Family history of breast cancer, or BRCA mutation without personal breast cancer, does not require restricting hormonal contraception.

Current or recent breast cancer is different. Hormonal contraception, including levonorgestrel IUD, is category four for current or recent breast cancer. If there has been no evidence of disease for at least five years, hormonal methods are category three. Copper IUD is category one after breast cancer treatment.

So the distinction is not "breast cancer word equals no contraception."

It is family history or mutation without personal disease versus hormone-sensitive cancer history.

Drug interactions

This is one of the most practically useful parts of the lecture.

Rifampin, rifabutin, and liver enzyme-inducing antiepileptic or antiretroviral drugs can reduce contraceptive steroid efficacy.

Combined hormonal contraception and progestin-only pills are generally category three with these drugs because of failure risk.

DMPA and levonorgestrel IUD can be used without concern for increased contraceptive failure, category one. The implant is category two with enzyme-inducing antiepileptics because efficacy may be susceptible, although it remains very effective overall.

The mechanism is steroid exposure.

If the method depends on low systemic steroid levels that can be metabolized faster, enzyme induction matters.

If the method is intrauterine or DMPA, much of the problem is bypassed.

Separate rifampin from ordinary antibiotics

Broad-spectrum antibiotics, antifungals, and antiparasitics other than rifamycins do not interfere with oral contraceptive efficacy in the pharmacokinetic evidence cited by ACOG. Tetracycline, doxycycline, ampicillin, metronidazole, quinolone antibiotics, and fluconazole do not behave like rifampin.

The resident reflex is: do not turn every antibiotic into rifampin.

Lamotrigine is a different kind of interaction.

Lamotrigine does not mainly threaten contraceptive efficacy. Estrogen-containing contraception reduces lamotrigine levels, and lamotrigine levels can rise during hormone-free intervals. Dose adjustment and coordination with neurology may be needed.

So the question is not only, will the medication make contraception fail?

Sometimes the contraception changes the medication.

Anticoagulation

All progestin-only methods are acceptable for women taking anticoagulants, category two. Warfarin is teratogenic, so reliable contraception matters. Anticoagulated patients may also have heavy menstrual bleeding or hemorrhagic ovarian cyst risk. A fifty two milligram levonorgestrel IUD can be especially useful because it provides reliable contraception and reduces menstrual blood loss. DMPA can reduce heavy bleeding and prevent ovarian cyst formation and rupture.

Combined hormonal methods may reduce bleeding too, but they add VTE concern and are considered case-by-case, generally category three in the relevant risk context.

Rehearse the model with cases

Case one.

A twenty five year old smoker asks for combined pills.

The tempting error is to say smoking means no combined pills. But smoking younger than thirty five is not the same as smoking at thirty five or older. She needs smoking counseling and cardiovascular risk assessment, but young smoking alone is not the same category as a thirty eight year old heavy smoker.

The lesson: age changes the smoking category.

Case two.

A thirty eight year old smokes twenty cigarettes per day and asks for the patch.

The patch is combined hormonal contraception. At age thirty five or older with fifteen or more cigarettes per day, combined hormonal contraception is category four.

The safer conversation is copper IUD, levonorgestrel IUD, implant, or another nonestrogen method depending on her goals.

The lesson: do not let a different estrogen delivery system hide the same contraindication.

Case three.

A patient says she has an MTHFR mutation and was told she can never use hormones.

The tempting error is to treat MTHFR as a thrombophilia equivalent.

The better response is to clarify. MTHFR mutations alone do not appear to increase VTE risk in the way that known thrombogenic mutations do, and testing MTHFR polymorphisms is not recommended for this decision. If she has antiphospholipid syndrome, factor five Leiden, prothrombin mutation, protein C deficiency, protein S deficiency, or antithrombin deficiency, that is a different conversation.

The lesson: similar sounding labels are not similar physiology.

Case four.

A patient with SLE wants contraception, and antiphospholipid antibody status is unknown.

Do not guess.

For estrogen, unknown antiphospholipid antibody status is treated as a high-risk state. Combined hormonal contraception is category four if antibodies are positive or unknown. Check the antibody status and offer safer alternatives while the risk is clarified.

The lesson: in lupus, the antibody result is not a detail. It changes the method.

Case five.

A breastfeeding patient is sixteen days postpartum and asks for the combined pill.

Combined hormonal contraception is category four in the first twenty one days postpartum regardless of breastfeeding. The safest answer is not, come back when breastfeeding stops. The safest answer is to offer immediate effective alternatives, especially progestin-only or nonhormonal methods as appropriate.

The lesson: early postpartum estrogen is a clotting problem first.

Case six.

A patient has migraine with visual symptoms before headache, lasting thirty minutes.

That is concerning for aura. Estrogen-containing contraception is category four. Progestin-only methods are not restricted by aura.

The lesson: headache history is not complete until aura is clarified.

Case seven.

A patient has renal transplant.

The resident asks one more question: is there graft failure?

If no graft failure, combined hormonal contraception is category two. If graft failure, it is category four. IUD initiation and continuation are handled differently but remain generally acceptable, with initiation category two and continuation category one in graft failure.

The lesson: transplant is not one category. Graft function changes the answer.

Case eight.

A patient takes carbamazepine.

The problem is enzyme induction and contraceptive steroid failure. DMPA and IUD methods are preferred because they bypass much of that interaction. Combined hormonal contraception and progestin-only pills are generally not recommended because of reduced efficacy.

The lesson: drug interactions are often failure-risk problems, not side-effect problems.

Case nine.

A patient takes lamotrigine.

The mechanism flips. Estrogen can reduce lamotrigine levels, and levels may rise during hormone-free intervals. The issue is seizure control and medication toxicity, not just contraception failure.

The lesson: sometimes the contraceptive changes the other drug.

Bring the whole lecture together

Medical eligibility is not a list of forbidden diagnoses.

It is risk reasoning.

Estrogen asks about clotting, arterial disease, aura, postpartum timing, smoking age, vascular disease, and hormone-sensitive cancer.

Progestin-only methods often become safer alternatives, but DMPA has its own cautions in selected vascular, diabetic, transplant, bone, and cardiovascular contexts.

IUD methods are powerful because they are highly effective and often avoid systemic estrogen risk, but anatomy, infection context, breast cancer history for levonorgestrel, and insertion timing still matter.

Drug interactions ask whether steroid exposure will fall, or whether the contraceptive changes the other medication.

The resident reflex is this:

Name the method.

Name the mechanism.

Name the category.

Offer the safer effective alternative.

Do not prescribe estrogen because the patient looks young.

Do not deny effective contraception because a label sounds scary.

The goal is not to say no.

The goal is to find the safest effective yes.